

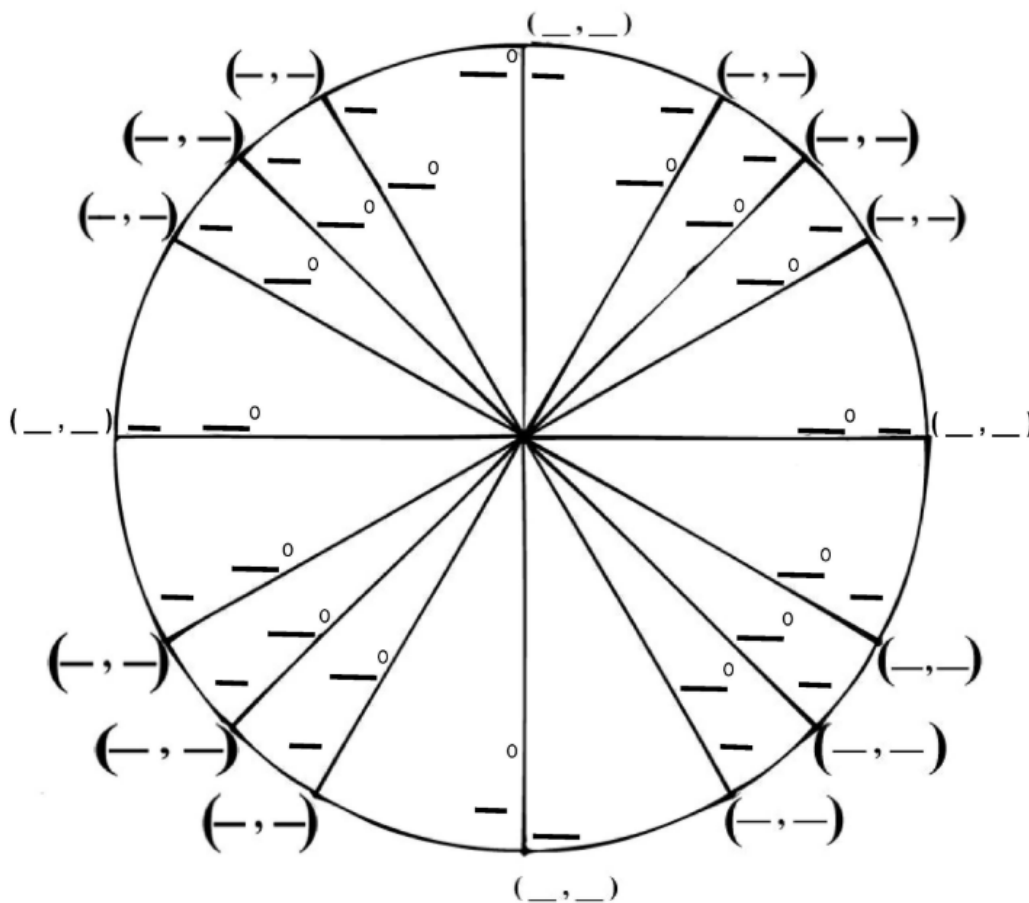
Name: _____

Exam # _____

Directions: There is a **3-hour time limit** on this exam. Please **fill in the bubble completely** for each multiple choice answer.

Potentially helpful information:

1. The quadratic formula (for solving $ax^2 + bx + c = 0$) is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
2. A blank unit circle that you can fill in:



Part I: True/False Statements: (2 points each) Choose True if the statement is a true identity, and False if the statement is false.

1. $\sin^2(\theta) + \cos^2(\theta) = 1$

☐ TRUE ☐ FALSE

2. $\csc^2(\theta) + 1 = \tan^2(\theta)$

☐ TRUE ☐ FALSE

3. $1 + \cot^2(\theta) = \csc^2(\theta)$

☐ TRUE ☐ FALSE

4. $\tan \theta = \frac{\sin \theta}{\cos \theta}$

☐ TRUE ☐ FALSE

5. $\sin(-\theta) = \sin(\theta)$

☐ TRUE ☐ FALSE

Part II: Multiple Choice Questions: (3 points each) Choose the best answer for each multiple choice question.

6. Find the equation of the line perpendicular to $y = -5x + 2$ that passes through the point $(-10, 1)$.

☐ $y = \frac{1}{5}x - \frac{16}{3}$

☐ $y = \frac{1}{5}x + 3$

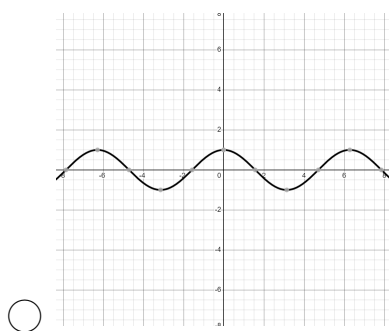
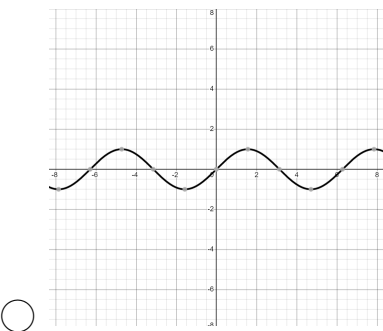
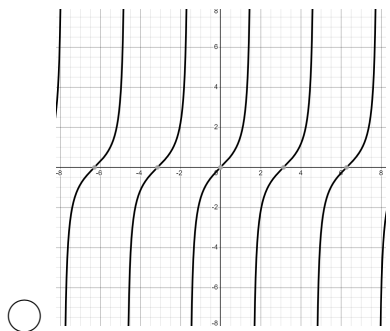
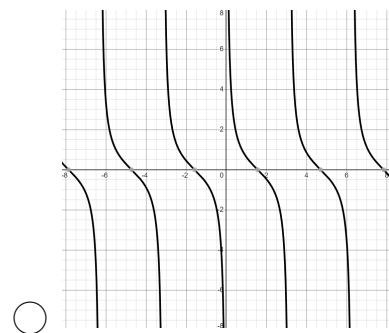
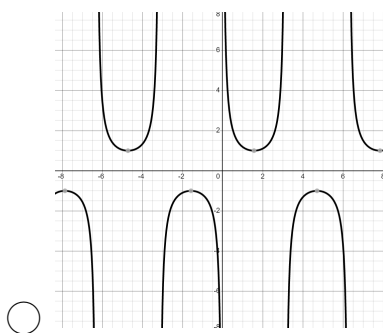
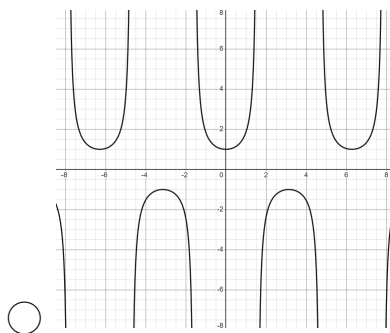
☐ $y = -5x - 12$

☐ $y = -\frac{1}{5}x + 10$

☐ $y = -\frac{1}{5}x$

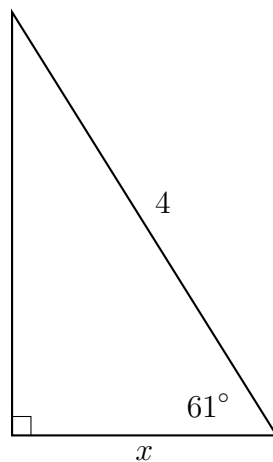
☐ $y = -5x$

7. Which of the following is the graph of $\cot(x)$?



8. Find an expression for the side x in the figure shown.

- ☐ $x = 4 \cos 61^\circ$
- ☐ $x = \frac{4}{\cos 61^\circ}$
- ☐ $x = 4 \sin 61^\circ$
- ☐ $x = \frac{4}{\sin 61^\circ}$
- ☐ $x = 4 \tan 61^\circ$
- ☐ $x = \frac{4}{\tan 61^\circ}$



9. Which of the following is the domain and range of $\sin^{-1}(x)$

- ☐ Domain: $[-1, 1]$, Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- ☐ Domain: $(-\infty, \infty)$, Range: $[-1, 1]$
- ☐ Domain: $[-1, 1]$, Range: $[0, \pi]$
- ☐ Domain: $(-\infty, \infty)$, Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- ☐ Domain: $[-1, 1]$, Range: $(-\infty, \infty)$
- ☐ Domain: $[0, 2\pi]$, Range: $(-\infty, \infty)$

10. What is $\tan^{-1}(-1)$?

☐ $\frac{\pi}{4}$

☐ $\frac{3\pi}{4}$

☐ $-\frac{\pi}{4}$

☐ $\frac{7\pi}{4}$

☐ $-\pi$

☐ π

11. Which of the following represents first, a horizontal stretch by a factor of 4, then second, a vertical shift down by 3, and lastly a vertical stretch by 2, if the original function is $f(x) = \sin(x)$?

☐ $g(x) = \sin(4x)$

☐ $g(x) = \sin(4x) + 3$

☐ $g(x) = 4\sin(2x) - 3$

☐ $g(x) = 2\sin(4x) - 3$

☐ $g(x) = 2(\sin(4x) - 3)$

☐ $g(x) = 4(\sin(2x) + 3)$

12. Find an expression for the angle θ in the figure shown.

☐ $\theta = \sin^{-1}\left(\frac{14}{5}\right)$

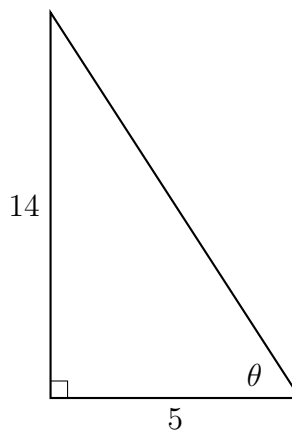
☐ $\theta = \sin^{-1}\left(\frac{5}{14}\right)$

☐ $\theta = \cos^{-1}\left(\frac{14}{5}\right)$

☐ $\theta = \cos^{-1}\left(\frac{5}{14}\right)$

☐ $\theta = \tan^{-1}\left(\frac{14}{5}\right)$

☐ $\theta = \tan^{-1}\left(\frac{5}{14}\right)$



13. Consider $f(\theta) = \tan(\theta)$. Which of the following corresponds to an asymptote on the graph of f ?

☐ $\theta = \pi$

☐ $\theta = \frac{\pi}{4}$

☐ $\theta = \frac{7\pi}{3}$

☐ $\theta = -\frac{13\pi}{6}$

☐ $\theta = \frac{3\pi}{2}$

☐ There are no asymptotes on the graph of f .

14. Let $f(x) = 2 \cos \left(8 \left(x + \frac{1}{3} \right) \right) - 1$. What is the period of the graph $y = f(x)$?

- ☐ Period = π
- ☐ Period = $\frac{\pi}{4}$
- ☐ Period = 2π
- ☐ Period = $\frac{\pi}{2}$
- ☐ Period = $\frac{\pi}{3}$
- ☐ Period = 4π

15. What are ALL solutions in the interval $0 \leq \theta < 2\pi$ to the equation:

$$2 \cos(\theta) = -1?$$

- ☐ $\theta = \frac{\pi}{3}$
- ☐ $\theta = -\frac{\pi}{3}$
- ☐ $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$
- ☐ $\theta = \frac{4\pi}{3}$ and $\theta = \frac{5\pi}{3}$
- ☐ $\theta = \frac{\pi}{3}$ and $\theta = -\frac{\pi}{3}$
- ☐ None of the above.

Part II: Free Response Questions: *Neatly write solutions for the following problems. All work must be shown in order to receive full credit.*

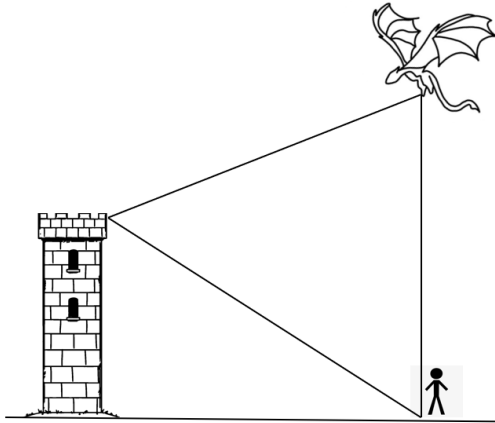
16. (6 points) Algebraically verify the following identity:

$$\cos(\theta) \cot(\theta) = \csc(\theta) - \sin(\theta)$$

17. (6 points) Find the exact value:

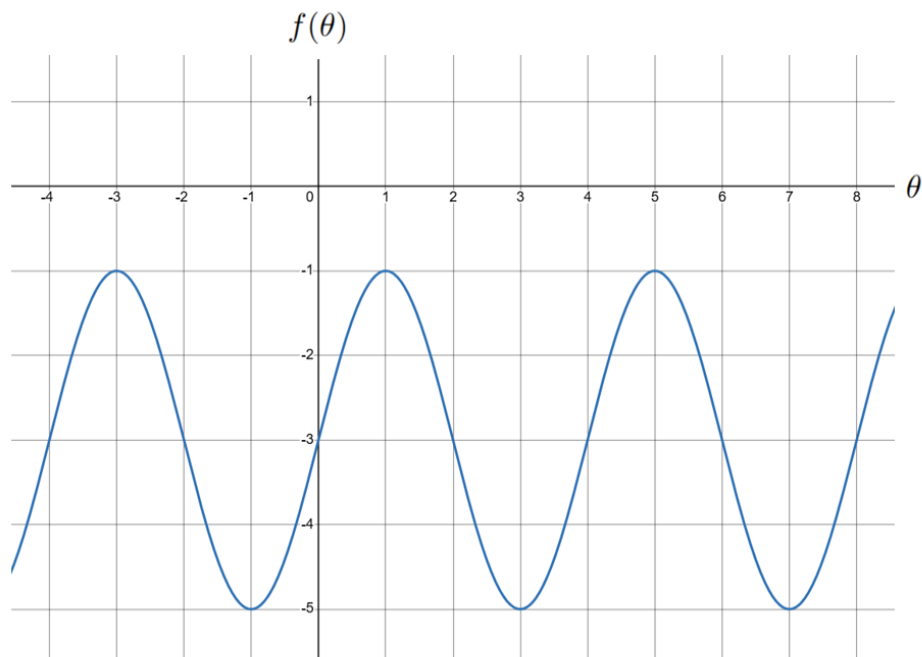
$$\cos \left(\sin^{-1} \left(\frac{4}{7} \right) \right)$$

18. (8 points) A guard stands on top of a castle tower that is 30 meters tall. He sees a villager on the ground and a dragon flying directly above the villager. The angle of depression to the villager is 30° , and the angle of elevation to the dragon is 60° . Find how high the dragon is flying. Show all your work, simplify your answer and include units. Note the picture is not drawn to scale.



19. Complete the following.

- a) (4 points) Write an equation for the function shown, expressing your answer as a sine function.



- b) (4 points) Find the following for the function $f(x) = 3 \sin(5x - 15) + \frac{1}{6}$.

- Amplitude: _____
- Period: _____
- Midline: _____
- Horizontal Shift: _____

20. Find ALL solutions to the following equations. If an interval is specified, restrict your answers to that interval. Show your work.

a) (6 points)

$$2 \sin\left(\frac{\pi}{4}\theta\right) = 1$$

b) (8 points)

$$-2 \sin^2 \theta + \cos \theta + 1 = 0 \quad \text{for } 0 \leq \theta < \pi$$